

Zephylin Dusengimana

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EDUCATION

Oklahoma Christian University <i>Master of Science in Computer Science GPA: 4.0/4.0</i>	Oklahoma, USA Aug. 2024 – Dec. 2025
Kyungdong University <i>Bachelor of Science in Computer Engineering GPA: 4.38/4.5 Top Student</i>	Goseong, South Korea Sep. 2020 – May 2024

PROFESSIONAL EXPERIENCE

AI Software Engineer <i>AfiaSync</i> - Built an end-to-end clinical documentation pipeline using Azure OpenAI (GPT-4o) and Azure Speech Services, reducing practitioner documentation time from 15 min to 2 min per patient encounter. - Engineered a versioned prompt framework with JSON schema enforcement and hallucination guardrails, increasing SOAP note quality scores from 40% to 80% on a standardized clinical rubric. - Designed a Python LLM-as-a-judge evaluation pipeline benchmarking AI outputs across 30 clinical scenarios, cutting prompt iteration cycle time from hours to under 60 seconds.	Jan. 2026 – Present Atlanta, GA
AI Software Engineer Intern <i>AfiaSync</i> - Contributed on backend services development using .NET 8 and Azure Cosmos DB with multi-tenant data isolation, enforcing HIPAA-compliant partition-key segregation. - Automated end-to-end testing using Playwright, reducing manual testing workload by 95%.	June 2025 – Dec. 2025 Remote
Machine Learning Engineer Intern <i>HighView Tech Group</i> - Engineered a real-time anomaly detection system for fuel monitoring, utilizing Python and SQL to process large-scale sensor data while maintaining system reliability and high performance. - Collaborated with engineers to integrate data pipelines into existing systems, using pandas and numpy for predictive modeling which improved data quality, observability, and operational reliability.	Jan 2024 – Apr. 2024 Kigali, Rwanda
Undergraduate Research Assistant <i>AI and Computer Vision Lab</i> Researched and developed a multi-scale object detection model using YOLOv11, enhancing accuracy by 25% across varied conditions and achieving 100% detection in real-world environmental monitoring scenarios.	Mar. 2022 – May 2024 Goseong, South Korea

PROJECTS

Agentic AI Shopping Assistant (Vesoko) <i>Python, LangGraph, MCP, FastAPI</i> - Architected a 15+ tool MCP server and LangGraph ReAct agent, building data pipelines to integrate AI models with a production e-commerce platform driving 35% conversion lift and reducing latency by 40% - Built an end-to-end AI ad campaign pipeline with GPT-4o for multilingual copy generation, AI image enhancement, and Meta Graph API integration, with automated performance insights and optimization recommendations - Designed RESTful microservices (FastAPI, Node.js/Express) with MongoDB Atlas Search, and Redis caching, deploying via CI/CD (Cloud Build) with structured logging and graceful degradation
Automatic Attendance Management System <i>Python, InsightFace, Redis</i> - Developed a facial recognition-based attendance system using InsightFace achieving 96% recognition accuracy - Designed an end-to-end data ingestion and processing pipeline to capture, process, and store facial recognition embeddings, enabling near real-time analytics across 5,000+ records.

TECHNICAL SKILLS

Programming Languages: Python, C#, Java, SQL, TypeScript
Backend & Integration: .NET 8, SpringBoot, RESTful APIs, RBAC
AI & ML: LangChain, LangGraph, MCP, PyTorch, TensorFlow, Scikit-Learn, RAG, Agentic AI, Transformers
Data Engineering & Analytics: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Jupyter Notebook, ETL/ELT pipelines
Databases: PostgreSQL, MongoDB, Redis, Azure Cosmos DB
Cloud & DevOps: Azure, AWS, Docker, Kubernetes, Git, CI/CD (GitHub Actions), Agile (Jira)